# Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matrix multiplication

128 × 128 · 128 × 128 · f64

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0009641651936365** | 0.9952760966204274 – 1.0073507188808772 |

## Matrix multiplication

128 × 128 · 128 × 128 · f32

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0004347259053168** | 0.9980513997445168 – 1.0019336476416016 |

# Version comparison / paired processes

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## Matrix multiplication

256 × 256 · 256 × 256 · f64

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.001448242726027** | 0.9985006212294634 – 1.0042017208478298 |

## Matrix multiplication

256 × 256 · 256 × 256 · f32

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.999291967226579** | 0.9974740323731636 – 1.0009405573441617 |

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## Independent matrix operands

128 × 128 · 128 × 128 · f64

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.9995639040163377** | 0.9955594464602143 – 1.003105490635505 |

## Independent matrix operands

128 × 128 · 128 × 128 · f32

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.001321604991767** | 0.9934858354794728 – 1.0058788947677837 |

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## Cholesky factorization

128 × 128 · f64

**Observed time increase**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0105549175378568** | 1.0039669636470052 – 1.0137466886231832 |

## Cholesky factorization

256 × 256 · f64

**Observed time reduction**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.6790232533939629** | 0.6733664516997802 – 0.6847512849206271 |

# Version comparison / paired processes

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## Cholesky factorization

512 × 512 · f64

**Observed time reduction**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **0.5090629428301201** | 0.5064715577820311 – 0.5154443163666095 |

## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

**Observed time increase**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0058258237116304** | 1.0012331107861963 – 1.019572302490545 |

# Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

## Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0005125276025437** | 0.9991821128070911 – 1.0050150348855331 |

## Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

**Timing direction unclear**

| CURRENT / BASELINE | 95% INTERVAL |
|---|---|
| **1.0015813276827221** | 0.9912324492979719 – 1.061308636493387 |